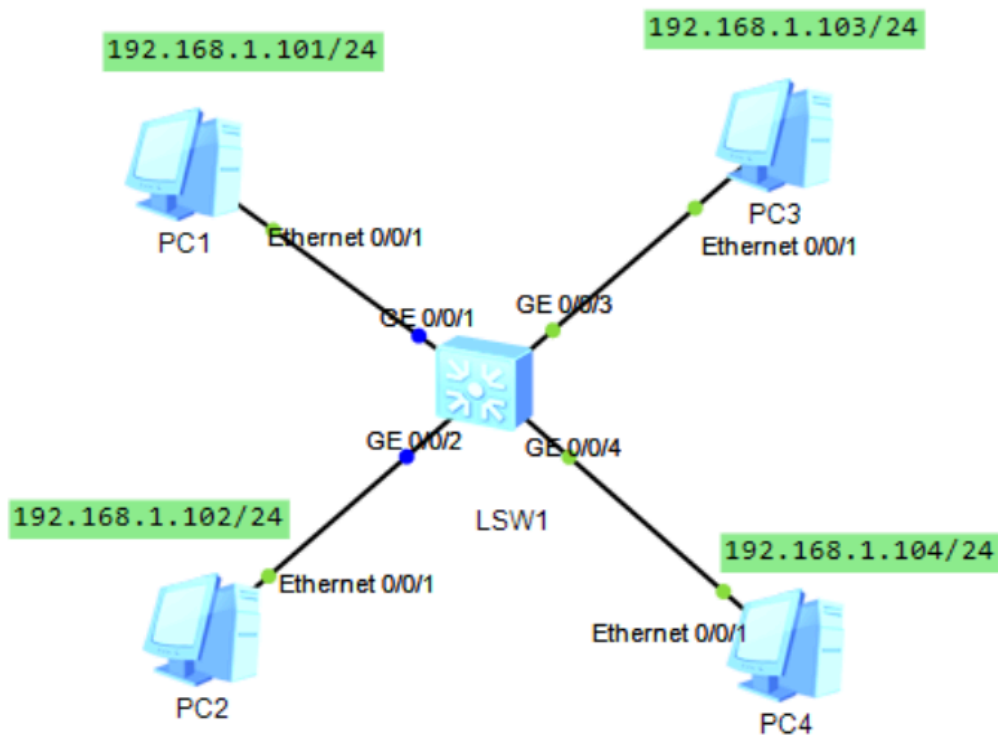


Lab13

13.1



```
PC>ping 192.168.1.102

Ping 192.168.1.102: 32 data bytes, Press Ctrl_C to break
From 192.168.1.102: bytes=32 seq=1 ttl=128 time=62 ms
From 192.168.1.102: bytes=32 seq=2 ttl=128 time=63 ms
From 192.168.1.102: bytes=32 seq=3 ttl=128 time=47 ms
From 192.168.1.102: bytes=32 seq=4 ttl=128 time=32 ms
From 192.168.1.102: bytes=32 seq=5 ttl=128 time=47 ms

--- 192.168.1.102 ping statistics ---
 5 packet(s) transmitted
 5 packet(s) received
 0.00% packet loss
 round-trip min/avg/max = 32/50/63 ms
```

20 42.843000	HuaweiTechno_a1:1b:: Broadcast	ARP	60 Who has 192.168.1.102? Tell 192.168.1.101
21 42.906000	HuaweiTechno_80:4c::	ARP	60 192.168.1.102 is at 54:89:98:80:4c:68
22 42.921000	192.168.1.101	192.168.1.102	ICMP 74 Echo (ping) request id=0xaa18, seq=1/256, ttl=128 (reply in 23)
23 42.960000	192.168.1.102	192.168.1.101	ICMP 74 Echo (ping) reply id=0xaa18, seq=1/256, ttl=128 (request in 22)
24 43.070000	HuaweiTechno_ad32::	Spanning-tree-(for...) STP	119 MST: Root = 32768/0/4c:1f:cc:ad:32:03 Cost = 0 Port = 0x0001
25 43.984000	192.168.1.101	192.168.1.102	ICMP 74 Echo (ping) request id=0xab18, seq=2/512, ttl=128 (reply in 26)
26 44.031000	192.168.1.102	192.168.1.101	ICMP 74 Echo (ping) reply id=0xab18, seq=2/512, ttl=128 (request in 25)
27 45.046000	192.168.1.101	192.168.1.102	ICMP 74 Echo (ping) request id=0xac18, seq=3/768, ttl=128 (reply in 28)
28 45.078000	192.168.1.102	192.168.1.101	ICMP 74 Echo (ping) reply id=0xac18, seq=3/768, ttl=128 (request in 27)
29 46.100000	HuaweiTechno_ad32::	Spanning-tree-(for...) STP	119 MST: Root = 32768/0/4c:1f:cc:ad:32:03 Cost = 0 Port = 0x0001
30 46.093000	192.168.1.101	192.168.1.102	ICMP 74 Echo (ping) request id=0xad18, seq=4/1024, ttl=128 (reply in 31)
31 46.125000	192.168.1.102	192.168.1.101	ICMP 74 Echo (ping) reply id=0xad18, seq=4/1024, ttl=128 (request in 30)
32 47.156000	192.168.1.101	192.168.1.102	ICMP 74 Echo (ping) request id=0xae18, seq=5/1280, ttl=128 (reply in 33)
33 47.167000	192.168.1.102	192.168.1.101	ICMP 74 Echo (ping) reply id=0xae18, seq=5/1280, ttl=128 (request in 32)

```
PC>ping 192.168.1.104

Ping 192.168.1.104: 32 data bytes, Press Ctrl_C to break
From 192.168.1.101: Destination host unreachable
From 192.168.1.101: Destination host unreachable
From 192.168.1.101: Destination host unreachable
From 192.168.1.101: Destination host unreachable
From 192.168.1.101: Destination host unreachable

--- 192.168.1.104 ping statistics ---
 5 packet(s) transmitted
 0 packet(s) received
100.00% packet loss
```

7 11.750000	HuaweiTechno_a1:1b:: Broadcast	ARP	60 Who has 192.168.1.104? Tell 192.168.1.101
8 12.734000	HuaweiTechno_a1:1b:: Broadcast	ARP	60 Who has 192.168.1.104? Tell 192.168.1.101
9 13.430000	HuaweiTechno_a1:1b:: Spanning-tree-(for...) STP		119 P2T: Root = 32768/0/4c:1f:1c:ad:32:00 cost = 0 Port = 0x8001
10 13.750000	HuaweiTechno_a1:1b:: Broadcast	ARP	60 Who has 192.168.1.104? Tell 192.168.1.101
11 14.750000	HuaweiTechno_a1:1b:: Broadcast	ARP	60 Who has 192.168.1.104? Tell 192.168.1.101
12 15.641000	HuaweiTechno_a1:1b:: Spanning-tree-(for...) STP		119 P2T: Root = 32768/0/4c:1f:1c:ad:32:00 cost = 0 Port = 0x8001
13 15.766000	HuaweiTechno_a1:1b:: Broadcast	ARP	60 Who has 192.168.1.104? Tell 192.168.1.101

测试	问题	ONLY VLAN 1 pkt: unicast/broadcast	ONLY VLAN 1 哪 些节点收到 包 (PC1~PC4)	配置了 VLAN12 和 VLAN34 pkt: unicast/broadcast	配置了 VLAN12 和 VLAN34 哪 些节点收到 包 (PC1~PC4)
PC1 "ping" PC2	PC1 发送 ARP 请 求?	broadcast	PC2,3,4	broadcast	PC2
	PC2 发送 ARP 回 复?	unicast	PC1	unicast	PC1
	PC1 发送 ICMP echo 请 求?	unicast	PC2	unicast	PC2
	PC2 发送 ICMP echo 回 复?	unicast	PC1	unicast	PC1
PC1 "ping" PC4	PC1 发送 ARP 请 求?	broadcast	PC2,3,4	broadcast	PC2

测试	问题	ONLY VLAN 1 pkt: unicast/broadcast	ONLY VLAN 1 哪 些节点收到 包 (PC1~PC4)	配置了 VLAN12 和 VLAN34 pkt: unicast/broadcast	配置了 VLAN12 和 VLAN34 哪 些节点收到 包 (PC1~PC4)
	PC4 发送 ARP 回 复?	unicast	PC1	-	-
	PC1 发送 ICMP echo 请 求?	unicast	PC4	-	-
	PC4 发送 ICMP echo 回 复?	unicast	PC1	-	-

13.2 (1)

Q1:

before

```
[LSW1]display mac-address
[LSW1]display ip routing-table
Route Flags: R - relay, D - download to fib
-----
Routing Tables: Public
    Destinations : 2          Routes : 2

Destination/Mask    Proto   Pre  Cost           Flags NextHop         Interface
-----
      127.0.0.0/8    Direct   0     0             D    127.0.0.1         InLoopBack0
      127.0.0.1/32    Direct   0     0             D    127.0.0.1         InLoopBack0

[LSW1]
```

after

```
[LSW1]display mac-address
MAC address table of slot 0:
-----
MAC Address      VLAN/      PEVLAN CEVLAN Port      Type      LSP/LSR-ID
                  VSI/SI
-----
5489-98a1-1ba6 12          -      -      GE0/0/1      dynamic   0/-
5489-9880-4c68 12          -      -      GE0/0/2      dynamic   0/-
-----
Total matching items on slot 0 displayed = 2

[LSW1]display ip routing-table
Route Flags: R - relay, D - download to fib
-----
Routing Tables: Public
          Destinations : 4          Routes : 4

Destination/Mask    Proto    Pre  Cost      Flags NextHop          Interface
-----
      127.0.0.0/8    Direct   0     0          D    127.0.0.1        InLoopBack0
      127.0.0.1/32    Direct   0     0          D    127.0.0.1        InLoopBack0
     192.168.1.0/24    Direct   0     0          D    192.168.1.1      Vlanif12
     192.168.1.1/32    Direct   0     0          D    127.0.0.1        Vlanif12
```

MAC 地址表: 增加了 Vlanif12 接口的 MAC 地址

路由表: 增加了 192.168.1.0/24 的直连路由

Q2. VLAN 接口 12 的路由条目

- **Proto:** Direct
- **Pre:** 0
- **Cost:** 0
- **Flags:** D
- **NextHop:** 192.168.1.1
- **Interface:** Vlanif12

Q3

PC1:

```
PC>ipconfig /renew

IP Configuration

Link local IPv6 address.....: fe80::5689:98ff:fea1:1ba
IPv6 address.....: :: / 128
IPv6 gateway.....: ::
IPv4 address.....: 192.168.1.252
Subnet mask.....: 255.255.255.0
Gateway.....: 192.168.1.1
Physical address.....: 54-89-98-A1-1B-A6
DNS server.....:
```

PC2:

```
PC>ipconfig /renew

IP Configuration

Link local IPv6 address.....: fe80::5689:98ff:fe80:4c68
IPv6 address.....: :: / 128
IPv6 gateway.....: ::
IPv4 address.....: 192.168.1.251
Subnet mask.....: 255.255.255.0
Gateway.....: 192.168.1.1
Physical address.....: 54-89-98-80-4C-68
DNS server.....:
```

Q4. ARP 请求能到达 PC1 吗？

答案：不能

```
PC>ping 192.168.1.252

Ping 192.168.1.252: 32 data bytes, Press Ctrl_C to break
From 192.168.1.103: Destination host unreachable
From 192.168.1.103: Destination host unreachable
From 192.168.1.103: Destination host unreachable
From 192.168.1.103: Destination host unreachable
From 192.168.1.103: Destination host unreachable

--- 192.168.1.252 ping statistics ---
    5 packet(s) transmitted
    0 packet(s) received
 100.00% packet loss
```

16 33.797000	HuaweiTechno_35:05:: Broadcast	ARP	60 who has 192.168.1.252? Tell 192.168.1.103
17 34.010000	HuaweiTechno_4d:32:: Spanning-tree-(for...	STP	119 HRT: Root = 32768/0/40-12/0/0/0/0/0 cost = 0 Port = 0x0003
18 34.813000	HuaweiTechno_35:05:: Broadcast	ARP	60 who has 192.168.1.252? Tell 192.168.1.103
19 35.813000	HuaweiTechno_35:05:: Broadcast	ARP	60 who has 192.168.1.252? Tell 192.168.1.103
20 36.797000	HuaweiTechno_35:05:: Broadcast	ARP	60 who has 192.168.1.252? Tell 192.168.1.103
21 36.920000	HuaweiTechno_4d:32:: Spanning-tree-(for...	STP	119 HRT: Root = 32768/0/40-12/0/0/0/0/0 cost = 0 Port = 0x0003
22 37.813000	HuaweiTechno_35:05:: Broadcast	ARP	60 who has 192.168.1.252? Tell 192.168.1.103

原因：PC3 在 VLAN 34，PC1 在 VLAN 12，ARP 广播不能跨 VLAN 传播。

Q5. ICMP 请求能到达 PC1 吗？

答案：不能

原因：ARP 失败，PC3 不知道 PC1 的 MAC 地址，无法发送 ICMP。

Q6. ICMP 回复能到达 PC3 吗？

答案：不能

原因：PC1 收不到 ICMP 请求，不会发送回复。

13.2(2)

Q1

```
[LSW1]display ip routing-table
Route Flags: R - relay, D - download to fib
-----
Routing Tables: Public
      Destinations : 6           Routes : 6

Destination/Mask    Proto   Pre  Cost   Flags NextHop         Interface
-----
127.0.0.0/8        Direct  0    0       D   127.0.0.1       InLoopBack0
127.0.0.1/32       Direct  0    0       D   127.0.0.1       InLoopBack0
192.168.1.0/24     Direct  0    0       D   192.168.1.1     Vlanif12
192.168.1.1/32     Direct  0    0       D   127.0.0.1       Vlanif12
192.168.2.0/24     Direct  0    0       D   192.168.2.1     Vlanif34
192.168.2.1/32     Direct  0    0       D   127.0.0.1       Vlanif34
```

新增了 192.168.2.0/24 的路由

```
PC>ping 192.168.1.252

Ping 192.168.1.252: 32 data bytes, Press Ctrl_C to break
From 192.168.1.252: bytes=32 seq=1 ttl=127 time=47 ms
From 192.168.1.252: bytes=32 seq=2 ttl=127 time=31 ms
From 192.168.1.252: bytes=32 seq=3 ttl=127 time=47 ms
From 192.168.1.252: bytes=32 seq=4 ttl=127 time=47 ms
From 192.168.1.252: bytes=32 seq=5 ttl=127 time=62 ms

--- 192.168.1.252 ping statistics ---
 5 packet(s) transmitted
 5 packet(s) received
 0.00% packet loss
 round-trip min/avg/max = 31/46/62 ms
```

Q7. PC3 会发送 ARP 请求吗？目标 IP 是什么？

会，目标ip是192.168.2.1

19 40.656000	HuaweiTechno_35:05:: Broadcast	ARP	60 Who has 192.168.2.1? Tell 192.168.2.103
20 40.672000	HuaweiTechno_ad:32:: HuaweiTechno_35:05:: ARP	60 192.168.2.1 is at 4c:1f:cc:ad:32:03	

Q8. PC1 会收到 ARP 请求吗？目标 IP 是什么？

不会收到

Q9. ICMP 请求能到达 PC1 吗？

能

22 31.765000	192.168.2.103	192.168.1.252	ICMP	74 Echo (ping) request id=0xaa35, seq=1/256, ttl=127 (reply in 23)
--------------	---------------	---------------	------	--

Q10. ICMP 回复能到达 PC3 吗？

能

23 31.781000	192.168.1.252	192.168.2.103	ICMP	74 Echo (ping) reply id=0xaa35, seq=1/256, ttl=128 (request in 22)
--------------	---------------	---------------	------	--